

Ordinary Banach–Mazur rigidity of von Neumann algebras

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October 7, 2026

Abstract

Using OpenAI’s all-algebra bounded Hochschild vanishing theorem, we derive local ordinary complex Banach–Mazur rigidity for every von Neumann algebra and its canonical predual. The threshold depends on the algebra; the conclusion is Jordan $*$ -isomorphism. Roydor announced the general cohomological implication in 2020, while his published Theorem 1.2 assumes separable predual. We give an explicit extension of his geometric argument using deformation stability on two fixed source algebras and reconstruction of odd finite type-I summands from full abelian projections. This avoids separability, moving-corner perturbation constants, and enumeration of infinite cardinal degrees. The cohomological input and geometric machinery are attributed external results; no new solution of the underlying cohomology problem or universal numerical threshold is claimed.

1 Statement, inputs and attribution

All Banach spaces and linear maps are complex. Define

$$d_{\text{BM}}(X, Y) = \inf_T \|T\| \|T^{-1}\|,$$

where T ranges over bounded bijective complex-linear maps with bounded inverse; the infimum is infinity when there is no such map. The literal formula gives $d_{\text{BM}}(0, 0) = 0$ and $d_{\text{BM}}(0, X) = \infty$ for $X \neq 0$. For a von Neumann algebra M , write M_* for its canonical predual, with $M = (M_*)^*$ isometrically. A Jordan $*$ -isomorphism is a bijective complex-linear map preserving involution and $x \circ y = (xy + yx)/2$.

Theorem 1. *For every complex von Neumann algebra M there is $\varepsilon_M > 0$ such that, for every complex von Neumann algebra N , either condition*

$$d_{\text{BM}}(M, N) < 1 + \varepsilon_M, \quad d_{\text{BM}}(M_*, N_*) < 1 + \varepsilon_M$$

implies that M and N are Jordan $$ -isomorphic. With this threshold, Jordan $*$ -isomorphism, algebra-space linear isometry, predual linear isometry, and the two displayed strict distance conditions are equivalent.*

The essential new input is the following attributed theorem.

Theorem 2 (OpenAI, family 295). *For every complex von Neumann algebra V and every $k \geq 2$, every bounded complex k -linear Hochschild cocycle $V^k \rightarrow V$ has a bounded complex $(k - 1)$ -linear primitive. Thus $H_b^k(V, V) = 0$.*

This is [3, Theorem 1.1], at the pinned repository version `adc7f124` (full identifier in the pinned bibliography link). The mathematical proof of that input has received independent source audits recorded in the supplement. We rely on its manuscript proof, not a claim that our follow-on result has been formalized. No fresh Lean kernel build of the upstream declarations was reproduced.

Roydor’s published reduction [1, Theorem 1.2] assumes *separable predual*, $H_b^2(M, M) = 0$, and closed $B^3(M, M)$ in $Z^3(M, M)$. Degree-three vanishing supplies the latter condition, since $B^3 = Z^3$ and Z^3 is a closed kernel. For separable-predual M , Theorem 1 is therefore an immediate consequence of that published theorem and Theorem 2. Roydor’s earlier author slides already announce the unrestricted conditional formulation [2, Theorem 2, PDF pp. 28–29]. Our contribution is the explicit fixed-source extension below and the consequence of the newly available cohomology input, not priority for that conditional formulation or invention of its geometric machinery.

The supplied complete 26-page publisher-formatted Roydor version, dated December 2020 and carrying the article’s DOI, was read in full. Its printed Theorem 1.2 and type-I argument must not be replaced by the less restricted slide statement. We use only Corollary 2.4, Proposition 2.13, Lemma 3.1 and Theorem 3.2 of that article, and its approximate-order estimate, Lemma 2.5. Their proofs have no separability hypothesis. In Theorem 3.2’s proof, the off-diagonal relation in the heading of Claim 4 should read $T(e_{ij}) \simeq g_{ij} + h_{ji}$, as used immediately above and in Claim 7. We do not use the article’s Theorem 1.1 or its compressed-map invocation in Lemma 4.3.

2 An involutive form of multiplication stability

For a complex C^* -algebra A , let $C_b^k(A, A)$ be all bounded complex k -linear maps, with the usual multilinear norm; $C_b^0(A, A) = A$. We use actual-image Hochschild cohomology, with

$$\begin{aligned} (df)(a_1, \dots, a_{k+1}) &= a_1 f(a_2, \dots, a_{k+1}) \\ &\quad + \sum_{j=1}^k (-1)^j f(a_1, \dots, a_j a_{j+1}, \dots, a_{k+1}) \\ &\quad + (-1)^{k+1} f(a_1, \dots, a_k) a_{k+1}, \end{aligned}$$

and $(db)(a) = ab - ba$ in degree zero. Thus $H_b^k = \ker d / \text{im } d$, with no closure in the denominator. Neither normality nor complete boundedness is required. The differential printed in [1, Sec. 4.2.1] omits the last adjacent merge and has an incorrect final sign; the standard differential above is the intended one, agrees with family 295, and is used throughout this paper.

The following is an involutive version of the classical multiplication stability mechanism of Johnson and Raeburn–Taylor [5, 6]; see also the symmetry-preserving iteration in [4, Proposition 3.2]. We include the argument to make its field, involution and fixed-algebra constants explicit.

Lemma 3. *Let A be a fixed complex unital C^* -algebra with $H_b^2(A, A) = H_b^3(A, A) = 0$. There is $\delta_A > 0$ such that every bounded associative complex bilinear product μ on A satisfying*

$$\|\mu - m_A\| \leq \delta_A, \quad \mu(x, y)^* = \mu(y^*, x^*)$$

admits an invertible bounded complex-linear $$ -preserving map Φ with $\|\Phi - \text{id}\| < 1/2$ and $\Phi\mu(x, y) = \Phi(x)\Phi(y)$. If μ has the original unit, Φ preserves it.*

Proof. Put $Z^k = \ker(d : C_b^k \rightarrow C_b^{k+1})$. Since $H_b^3 = 0$, the map $d : C_b^2/Z^2 \rightarrow Z^3$ is a bounded bijection of Banach spaces. Since $H_b^2 = 0$, $d : C_b^1 \rightarrow Z^2$ is onto. The open mapping theorem gives constants $K, L \geq 1$ such that each $f \in C_b^2$ has $z \in Z^2$ with $\|f - z\| \leq K\|df\|$, and each $z \in Z^2$ has

a primitive h with $\|h\| \leq L\|z\|$. Choose K, L strictly larger than the respective quotient inverse bounds, to accommodate nonattained infima; for a zero right side choose the exact zero-error solution. A bounded linear selection is unnecessary.

Define conjugate-linear isometric involutions, for $n \geq 1$, by

$$S_n f(a_1, \dots, a_n) = (-1)^{(n-1)(n-2)/2} f(a_n^*, \dots, a_1^*)^*.$$

Reversing the differential's terms gives $dS_n = S_{n+1}d$. Averaging z with $S_2 z$, and its primitive with $S_1 h$, preserves the preceding bounds when $f = S_2 f$. The S_1 -fixed maps preserve the involution; S_2 -fixed bilinear maps satisfy the product symmetry in the statement.

Write $\mu = m_A + \Delta$, $\delta = \|\Delta\|$. Associativity gives

$$d\Delta(x, y, z) = \Delta(\Delta(x, y), z) - \Delta(x, \Delta(y, z)), \quad \|d\Delta\| \leq 2\delta^2.$$

Choose symmetric z, h with

$$dh = z, \quad \|\Delta - z\| \leq 2K\delta^2, \quad \|h\| \leq L(\delta + 2K\delta^2).$$

For $g = \text{id} + h$ and $b = g^{-1}$, transport the product to $\nu(x, y) = g\mu(bx, by)$. Its exact defect is

$$(\nu - m_A)(x, y) = [(\Delta - dh) + h\Delta - m_A(h\cdot, h\cdot)](bx, by).$$

If $\delta \leq 1/(2K)$ and $\delta \leq 1/(4L)$, then $\|h\| \leq 2L\delta \leq 1/2$, $\|b\| \leq 2$, and

$$\|\nu - m_A\| \leq D\delta^2, \quad D = 8K + 8L + 16L^2.$$

Transport preserves associativity and the involution. Take $\delta_A = \min\{1/(2K), 1/(16L), 1/(2D)\}$ and iterate, using the original differential of the fixed algebra A at every stage. Then $\delta_n \leq 2^{-n}\delta_0$ and $\sum_n \|h_n\| \leq 4L\delta_0 \leq 1/4$. The ordered products $\Phi_n = g_{n-1} \cdots g_0$ converge in operator norm and satisfy $\|\Phi - \text{id}\| \leq \exp(4L\delta_0) - 1 < 1/2$. Thus Φ is invertible and $*$ -preserving. Passing to the limit in $\Phi_n \mu(x, y) = \mu_n(\Phi_n x, \Phi_n y)$ proves the conjugacy. An onto algebra isomorphism sends a unit to a unit, proving the last assertion when both original products have unit 1. $\square$

3 Rigidity on a fixed algebra whose unit halves

Say that a von Neumann algebra *halves* if its unit is the sum of two orthogonal equivalent projections. Equivalently, its finite type-I homogeneous summands have even degrees only; properly infinite summands and type-II summands also halve. This is the projection-halving criterion used in Roydor's proof of Theorem 3.2, with the standard structure theory [7, 8]. All infinite cardinal dimensions are included.

Lemma 4. *Let V be a fixed von Neumann algebra which halves and satisfies $H_b^2(V, V) = H_b^3(V, V) = 0$. There is $\eta_V > 0$ such that every von Neumann algebra W with $d_{\text{BM}}(V, W) < 1 + \eta_V$ is Jordan $*$ -isomorphic to V .*

Proof. For a sufficiently small distance excess, Roydor's Corollary 2.4 supplies a unital self-adjoint linear isomorphism $T : V \rightarrow W$ with $\max\{\|T\|, \|T^{-1}\|\} \leq 1 + t$, where $t \rightarrow 0$ with that excess. His Theorem 3.2 and Lemma 3.1 provide central $p \in V, q \in W$ and a block isomorphism

$$F = T_p^q \oplus T_{1-p}^{1-q}$$

The first block nearly preserves multiplication; the second nearly reverses it. The estimates are

$$\begin{aligned} \|F^{-1}\| &\leq 1 + 988\sqrt{t}, \\ \max\{\|(T_p^q)^\vee\|, \|(T_{1-p}^{1-q})^\wedge\|\} &\leq 3147585\sqrt{t}. \end{aligned}$$

The superscripts denote the ordinary bilinear multiplicative and anti-multiplicative defects, respectively. Zero blocks are omitted. Put $y \diamond z = qyz + (1 - q)zy$ on W . Centrality of q makes this an associative unital C^* -product on the same normed $*$ -space, namely the original product on qW and the opposite product on $(1 - q)W$. The pulled-back product

$$\mu(x, y) = F^{-1}(F(x) \diamond F(y))$$

is associative, unital and involution-compatible, and

$$\|\mu - m_V\| \leq (1 + 988\sqrt{t})3147585\sqrt{t} \longrightarrow 0.$$

Apply Lemma 3 once to the fixed algebra V . Then $F\Phi^{-1}$ is a $*$ -isomorphism from V to $(W, \diamond)$, hence a Jordan $*$ -isomorphism to W with its original product. The constants are attached to V , not the moving projection p . $\square$

4 Full carriers and odd finite type-I summands

The central carrier $c(h)$ of a projection is the least central projection dominating it. We need the following quantitative observation.

Lemma 5. *Let $T : C \rightarrow D$ be a unital self-adjoint linear isomorphism of von Neumann algebras with both norms at most $1 + t$, for sufficiently small t . If $c_C(h) = 1$ and a projection $k \in D$ satisfies $\|T(h) - k\| \leq r(t) = 140\sqrt{t}$, then $c_D(k) = 1$.*

Proof. Roydor's Proposition 2.13 gives a center $*$ -isomorphism θ with $\|T(z) - \theta(z)\| \leq r(t)$ for central projections z . Put $w = c_D(k)$ and $z = \theta^{-1}(w)$. His Lemma 2.5 applied to T^{-1} and $k \leq w$ yields

$$T^{-1}(k) \leq T^{-1}(w) + \gamma(t)1, \quad \gamma(t) = 2(1 + t)\sqrt{2t + t^2}.$$

Both $\|h - T^{-1}(k)\|$ and $\|T^{-1}(w) - z\|$ are at most $(1 + t)r(t)$. Therefore

$$h - z \leq [2(1 + t)r(t) + \gamma(t)]1.$$

Choose t so that the bracket is less than one. Since z is central, $h(1 - z)$ is a projection; compressing the inequality by $1 - z$ forces it to be zero. Thus $h \leq z$, and $c_C(h) = 1$ forces $z = 1$, $w = 1$. The same argument applies to complements, since T is unital. $\square$

Lemma 6. *Let O be a fixed von Neumann algebra consisting of finite homogeneous type- I_n central summands with odd $n \geq 3$. Choose a projection e which has rank $n - 1$ on each such summand. Suppose $E = eOe$ satisfies $H_b^2(E, E) = H_b^3(E, E) = 0$. Then O is locally Jordan rigid for ordinary Banach–Mazur distance among von Neumann algebras, without a separability assumption.*

Proof. Both e and $1 - e$ have full central carrier; $(1 - e)O(1 - e)$ is abelian, and the fixed algebra E halves. These assertions follow by writing each finite homogeneous summand as its arbitrary abelian center tensored with M_n , choosing the diagonal rank- $n - 1$ projection, and taking the bounded central direct product.

Let $T : O \rightarrow D$ be a unital self-adjoint near isomorphism as in Lemma 4. Round $T(e)$ to a projection q using Roydor's Proposition 2.13. Then $1 - q$ is equally close to $T(1 - e)$. Lemma 5 gives $c_D(q) = c_D(1 - q) = 1$. Roydor's Lemma 3.1 gives near isomorphisms from E to qDq , and from $(1 - e)O(1 - e)$ to $(1 - q)D(1 - q)$, with both distortions tending to one. His Proposition 2.13 applied to the latter compression makes $(1 - q)D(1 - q)$ abelian. Its full carrier implies that D is type I. Lemma 4, applied to the single fixed algebra E , gives an exact Jordan $*$ -isomorphism $J : E \rightarrow qDq$.

For full projection corners, compression identifies their centers with the center of the original algebra. The center restriction of J therefore induces a $*$ -isomorphism $\beta : \mathcal{Z}(O) \rightarrow \mathcal{Z}(D)$. It preserves arbitrary existing joins: a Jordan $*$ -isomorphism and its inverse are positive order isomorphisms, hence normal. Let z_n be the source homogeneous degree- n central projection and put $w_n = \beta(z_n)$. The corner $w_n q D q$ is homogeneous of degree $n - 1$. This follows from the usual central homomorphism/anti-homomorphism decomposition of a von Neumann Jordan $*$ -isomorphism; either part preserves finite homogeneous degree.

Consequently $w_n q$ is the sum of $n - 1$ orthogonal equivalent abelian projections, each with central carrier w_n in D . The complementary projection $w_n(1 - q)$ is abelian and also has carrier w_n . Abelian projections with the same central carrier are equivalent [7, 8]. Thus w_n is the sum of n orthogonal equivalent abelian projections, and $w_n D$ is homogeneous of degree n . Its center is identified by β with that of $z_n O$, so finite homogeneous matrix classification gives $z_n O \cong w_n D$ as von Neumann algebras. Taking their bounded central direct product gives $O \cong D$. The centers can have arbitrary density. No measurable fiber decomposition or infinite-cardinal cancellation was used.

All constants were chosen for the fixed E , and the compression bounds depend only on the initial norm excess. Hence one threshold works for all D , all q and all finite odd degrees simultaneously. $\square$

5 Completion of the algebra and predual conclusions

Proof of Theorem 1. For $M \neq 0$, use its three fixed central summands

$$M = A \oplus P \oplus O,$$

where A is type I_1 , O contains the finite odd type- I_n summands with $n \geq 3$, and P contains everything else: finite even and all infinite type-I summands, and all type-II and type-III summands. Then P halves, and the projection e in Lemma 6 chooses the second fixed source algebra $E = eOe$. Theorem 2 applies to each of P, E , since both are von Neumann algebras. This is why the proof uses universal vanishing; we do not infer the noncentral corner's cohomology merely by applying a central restriction to M .

For sufficiently small ordinary distance, normalize the initial Banach isomorphism using Roydor's Corollary 2.4. Its Proposition 2.13 sends the three central projections to an orthogonal partition of 1_N , and Lemma 3.1 gives near isomorphisms from the fixed three source summands to the corresponding target summands, with uniform distortion bounds converging to one. The commutative pieces are $*$ -isomorphic by Proposition 2.13; Lemmas 4 and 6 handle P and O . Their finite direct sum is a Jordan $*$ -isomorphism $M \rightarrow N$. Omit zero summands. This includes arbitrary nonseparable centers and arbitrary infinite type-I cardinal dimensions.

Choose the deformation thresholds once for the at most two nonzero fixed algebras P, E . There are only finitely many subsequent center, compression, carrier and halving inequalities; all their norm error bounds tend to zero as the original distortion tends to one. Choosing a small positive ε_M to satisfy them proves the stated $\forall M \exists \varepsilon_M \forall N$ quantifiers. No universal constant is inferred.

An infimum-attaining map is unnecessary: a strict distance bound supplies some isomorphism with distortion strictly below that bound.

For the predual implication, if $S : M_* \rightarrow N_*$ is a bounded complex-linear isomorphism, its Banach adjoint $S^* : N \rightarrow M$ has inverse $(S^{-1})^*$ and exactly the same two norms. Taking infima gives

$$d_{\text{BM}}(M, N) \leq d_{\text{BM}}(M_*, N_*).$$

Thus the same algebra threshold suffices. Exact Jordan $*$ -isomorphisms are normal order isomorphisms and induce isometric canonical-predual maps. Conversely, predual isometries adjoint to algebra-space isometries, and Kadison’s isometry theorem gives Jordan $*$ -isomorphism; see also [9, Theorem 1.1]. This proves all equivalences. For $M = 0$, either strict distance inequality forces $N = 0$, so any positive threshold works. $\square$

6 Prior scope and verification limits

Ricard–Roydor’s unconditional theorem concerns *completely bounded* Banach–Mazur distance [4, Theorem A]; it cannot be used from small ordinary distance, since $d_{\text{BM}} \leq d_{cb}$ has the wrong direction for that inference. Its predual observation and conditional two-matrix-level version are also inherited [4, Remarks 3.4 and 3.7]. Roydor’s special ordinary-stable classes include those whose type-II₁ summand has property Γ , a Cartan MASA or McDuff absorption [1, Corollary 1.3]. Neither those cases nor exact predual duality are new contributions.

A fresh primary-source priority audit on October 7, 2026 checked the literature, equivalent formulations, companion manuscripts and current upstream version. No explicit duplicate full arbitrary-algebra consequence and proof was identified in that bounded search. This does not establish firstness. The general conditional formulation was already announced by Roydor; the present note gives a fixed-source proof extension and records a consequence of OpenAI’s newly public vanishing theorem. Opposite algebras explain the Jordan conclusion: ordinary Banach norm geometry need not determine the orientation of the associative product.

Independent mathematical source audits found no substantive gap in the pinned cohomology manuscript, and separate adversarial checks examined the fixed-source deformation, carrier and reconstruction arguments. Finite computations in the supplement are checks of proof mechanisms, not proofs of the infinite-dimensional input. Static authentication of the actual upstream Lean sources is distinguished from kernel verification: no fresh compilation/import or axiom probe was reproduced, and this Banach–Mazur consequence is not formalized here. Complete-package reviews and publication verification are recorded separately in the accompanying research records.

Disclosure. AI tools were used extensively in the research, drafting and verification. Automated adversarial reviews are internal checks, not conventional human peer review. This preprint has not undergone conventional human peer review or refereeing.

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